SHOCK WAVES ON THE HIGHWAY*

PAUL I. RICHARDS

Technical Operations, Inc., Arlington, Massachusetts

(Received July 1, 1955)

A simple theory of traffic flow is developed by replacing individual vehicles with a continuous 'fluid' density and applying an empirical relation between speed and density. Characteristic features of the resulting theory are a simple 'graph-shearing' process for following the development of traffic waves in time and the frequent appearance of shock waves. The effect of a traffic signal on traffic streams is studied and found to exhibit a threshold effect wherein the disturbances are minor for light traffic but suddenly build to large values when a critical density is exceeded.

THE THEORY OF TRAFFIC FLOW to be presented below rests on an idealized picture wherein many individual cars are replaced by a simple continuous distribution. In this way a number of interesting results can be obtained which exhibit many familiar features.

It is not claimed, however, that this simple picture is an accurate representation of the facts under all circumstances. One problem which it will *not* handle adequately, for example, is the length of waiting-lines at a toll booth. This quantity depends primarily on statistical fluctuations in traffic volume. The simple picture below gives only the result that a waiting-line must have a certain density of cars, but its length is indeterminate. Thus the simple theory fails in this case, but at least does not give an incorrect answer.

Previous analyses of problems of this nature have been reported by Reuschel,[1, 2] Pipes,[3, 4] Prager,[5] and Newell.[6] The first two authors consider a number of vehicles each of which is constrained to follow its predecessor according to some assumed 'rule' that relates its velocity and acceleration to the motions of its predecessors. These relations lead to coupled differential equations that, given initial conditions and the motion of the lead vehicle, can be solved by various methods to predict the motion of all vehicles. In practice, these theories have necessarily been applied to a relatively small number of vehicles. On the other hand, the present theory is valid only for a very large number of vehicles since the basic step of replacing traffic by a continuous distribution-function certainly loses validity for relatively few vehicles. Likewise the present theory, instead of proceeding from detailed assumptions concerning driver-behavior, endeavors to derive results with a minimum of hypotheses. As a result, it complements the earlier work by supplying broad over-all results but, be-

* This work was supported by the Operations Research Office of The Johns Hopkins University.

cause of its economy, is unable to yield detailed precise results for very low-density traffic.

Newell[6] reports on a kinetic theory model of traffic in which a collision is analogous to the passing of one car by another. This theory, as its author emphasizes, is primarily applicable to multi-lane high-speed highways under light traffic conditions. Specifically, the possibility of 'many-body collisions' analogous to 'traveling queues' is not included. These results, therefore, also complement the theory treated here, which, by contrast, applies primarily to heavy traffic where passing would be greatly impeded and indeed treats only the 'density waves' that develop under such conditions.

The work of Prager[5] treats a somewhat different problem. In its present form his theory concerns an essentially steady state on a road network and provides a model for the distribution of such steady streams among alternative routes. The theory presented here concerns essentially a single route under nonsteady conditions, but it might be of use in extending Prager's analysis to the dynamic case.

ANALYSIS

WE SHALL USE only two assumptions in what follows. The first might more properly be called a statement of the method. The second can be supported by empirical data.[7] The two assumptions are:

(a) Any distribution of cars can be replaced by a 'continuous' density function, $D(x, t)$ cars per mile say (where x measures position along the road and t is time).

(b) The velocity of individual cars depends on car-spacing in such a manner that, at any instant and at every point, cars in that vicinity have a velocity given by

$$V = a(b - D), \tag{1}$$

where a and b are suitable constants depending on the roadway. (Actually many of the results below can be generalized for arbitrary dependence of V on D, but relation (1) leads to especially simple results and is as well supported by data[7] as any other.)

Equation (1) will be simplified somewhat. First we note that $V=0$ when $D=b$. Thus b has the significance of a maximum possible density. It is approximately the density in a line of stalled traffic:

$$b = D_{\max}. \tag{2}$$

If we now introduce a fractional density, d, and maximum velocity, c, which are defined by

$$d(x,t) = D(x,t)/D_{\max}, \qquad c = ab, \tag{3}$$

then we may rewrite (1) in the form

$$V = c(1 - d). \tag{4}$$

Now these assumptions will be manipulated to obtain the results which they imply. If we think of the traffic as a compressible fluid of density d and fluid-velocity V, then we may write the fundamental equation of conservation of matter

$$\partial d/\partial t + \partial (Vd)/\partial x = 0. \tag{5}$$

Our 'traffic-fluid,' however, is unlike any fluid ever seen in nature in that there is a direct relation between density and velocity. The velocity was assumed to be given by (4) independent of all other considerations, and we shall see that (4) in a sense replaces Newton's law since the motions are completely determined by (4) and (5) alone.

Substituting (4) into (5), we obtain

$$\partial d/\partial t + c\,(1-2d)\,\partial d/\partial x = 0.$$

The general solution* of this equation may be found by standard methods and can be written

$$d = f[x + (2d-1)\,ct], \tag{6}$$

where f is an arbitrary function of the argument indicated. To be applicable to our problem, however, f must take only values lying between 0 and 1 as these are the only meaningful values for the fractional car-density d.

Equation (6) is an implicit relation for d and obviously cannot be solved in general terms to give an explicit expression. Fortunately this is not necessary. First, let $t=0$ in (6):

$$f(x) = d(x,0); \tag{7}$$

that is, $f(x)$ is the spatial distribution of fractional car density at zero time.

Now consider a fixed value of d in (6). If we thus concentrate on the motion of a point where d has a given fixed value d_0 we see that $x + (2d_0 - 1)\,ct$ must have a fixed value or in other words: *The points where* $d = d_0 =$ *constant travel with the velocity* $(1-2d_0)\,c$. This may be stated in another way as also follows directly from (6) and (7). *The graph of* $d(x,t)$ *versus* x *for any instant* t *can be derived from the graph of* $d(x,0)$ *by shearing the* d,x*-plane (uniformly) in such a manner that the line* $d=0$ *moves a distance* ct *to the right and the line* $d=1$ *an equal distance to the left, the line* $d=\frac{1}{2}$ *remaining fixed.* A simple example is shown in Fig. 1, which represents cars pulling away from their fellows on a clear road.

It must not be thought that this shearing motion of the *graph* in any way represents the motion of individual *cars*. We are studying the mo-

* In checking this solution by substitution, note that d appears on both sides of (6).

tions of *waves*; the individual cars at each point are traveling with velocity $c\,(1-d)$ and may either pass through a wave-crest or pile up behind it depending on the relative velocities involved.

The reader has probably noticed that the 'shearing rule' can lead to graphs which display physically impossible situations with multiple values

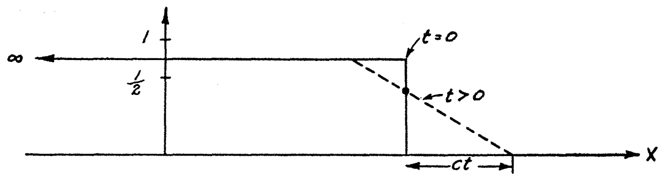

Figure 1

of d for a given x. One example of this is shown in Fig. 2, which represents a fast-moving line coming up behind a slow-moving line. This result is certainly incorrect as it stands, but we shall see that it is easily converted to the correct picture.

As is well-known in fluid mechanics, the end result of one volume of fluid overtaking another must be the formation of a *shock wave.* This is simply a density-discontinuity, which travels through the fluid in a manner

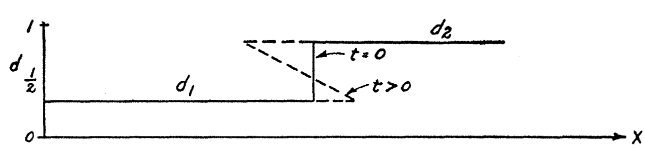

Figure 2

that is well determined by the laws of motion. If U is the velocity of the shock, conservation of matter (cars) gives the result:

$$d_1\,(V_1-U)=d_2\,(V_2-U), \tag{8}$$

where the subscripts denote the values of quantities immediately on either side of the shock-discontinuity.* Substituting for V_1 and V_2 from (4), the relation expressing the peculiar property of 'traffic fluid,' we find

$$U=c\,(1-d_1-d_2). \tag{9}$$

A fairly straightforward geometric argument will show that (9) is also equivalent to the following extension of the 'shearing rule': *If the graph-*

* To prove (8), transform coordinates to those of an observer moving with the shock-wave at velocity U. He sees cars traveling with velocities $V_1'=V_1-U$ and $V_2'=V_2-U$ on the two sides of the shock. The number of cars entering the shock from the one side, d_1V_1', per unit time must equal the number leaving the other side, d_2V_2', per unit time. Since the relation (8) is symmetric in the subscripts, it does not matter which side is denoted as '1.'

shearing process leads to multiple values of d for any x, the correct graph is obtained by connecting the top branch of the curve to the bottom branch by a vertical straight-line (shock-wave) so located that the areas of the loops cut off on either side of the line are equal (see Fig. 3). This rule also holds when the curve becomes multiply folded so that a shock line cuts off several loops on either side. It is usually simpler, however, actually to draw in the shocks

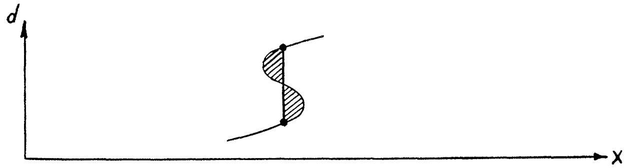

Figure 3

before this situation develops and erase the loops before continuing the shearing process.

These shock waves are perhaps the most characteristic feature of the model under discussion. Anyone who has driven on a high-speed road or parkway in heavy traffic has been made dramatically aware of existence of shocks in fact, as well as in theory. Whenever screeching tires and brakes bring us to a halt inches away from the car ahead, we have just

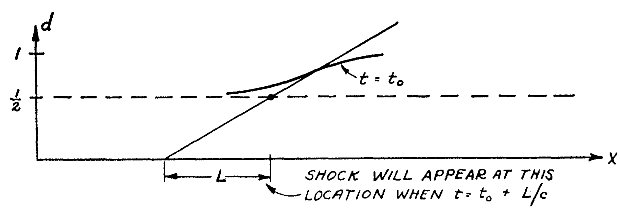

Figure 4

witnessed the passage of an especially strong shock. Examination of a few typical graphs of d versus x and their development in time by the shearing rules above will show that shock-waves always develop whenever the slope of the graph is positive ($\partial d/\partial x > 0$). In fact, as may be seen by some elementary geometric arguments using the 'shear rule,' the place and time where shocks will first appear can be predicted from the original graph by drawing tangents at points of maximum positive slope and noting the abscissae at which they cross the line $d = \frac{1}{2}$ (see Fig. 4).

Before considering some special examples in detail, a few further words on interpretation of the graphs may be in order. First, as already noted, the velocity of the cars is given by (4) and is not directly related to the shearing motion by which the waves develop. Examination of (4) will show that the graph of d at any instant is at the same time an *inverted* graph of V, the local car velocity. In fact, the distance of the wave below the line $d=1$ gives directly V/c, the ratio of the actual local velocity to the 'speed limit.'

Secondly, the slope of the graph is related to another quantity of interest, the acceleration of a car at a given place and time. In fact from (4) and (5) one finds

$$\text{acceleration} = dV/dt = \partial V/\partial t + V\,\partial V/\partial x = -c^2 d\,\partial d/\partial x.$$

Note that this relation can be interpreted as saying that a driver changes his speed at any instant in accord with conditions which he sees ahead (and behind?) as expressed by $\partial d/\partial x$. This result would not have been anticipated from our original assumption (4) alone (which says merely that he does not stop completely until $D \equiv D_{\max}$), but it is a consequence of (4) when combined with the conservation of cars, equation (5).

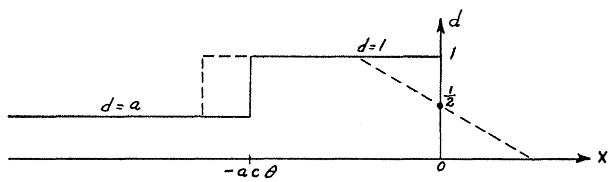

Figure 5

Finally, the number of cars passing a given point per unit time is often of interest. This traffic flow, F, is analogous to the *current* in hydrodynamics and is given by (DV) or

$$F = c\,D_{\max}\,d\,(1-d). \tag{10}$$

If c is in miles/hr and $D_{\max}$ in cars/mile, F will be in cars per hour. This flow varies with position even at any one instant of time, and it also can be discontinuous at a (moving) shock front. Note that the flow is the same for a density $1-d_0$ as it is for density d_0. This does not mean that the traffic conditions are the same in the two cases; in fact, the velocities are different and to the motorist the distinction is marked indeed, but to a robot traffic-counter the two cases are identical.

The first example we wish to consider is that due to a sudden interruption of an originally uniform stream of traffic. Let the density be initially given by $d=a$ at all points. This situation will continue until some constraint is imposed from without. Suppose something (an accident perhaps) forces the traffic at $x=0$ to stop for a time θ. During this time, the local velocity at $x=0$ must become zero. The density d therefore becomes unity and a shock develops and travels [backwards, see (9)] at a speed $U=-ac$, with the result that at the end of the stopping interval θ, the situation will be that shown by the solid graph in Fig. 5. Let us choose the instant illustrated in Fig. 5 as zero time (i.e., the original interruption occurred at time $-\theta$). At the instant $t=0$, the stalled traffic in Fig. 5 is suddenly released again. The development can now be followed by the 'shearing rule' (which also shows that neglect of the traffic which has pulled on ahead will not affect our results as long as we are interested only in the region $x \leq 0$).

During the initial moments after traffic flow resumes, the graph takes a form shown by the dotted curve in Fig. 5. Cars are still coming to a stop at the back edge where the original shock wave is still traveling up the line, but a 'starting wave' is rapidly overtaking it. In fact, the shock is traveling at a speed $(-ac)$ as given by (9) while the 'shear rule' shows that the

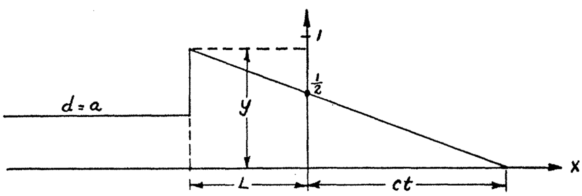

Figure 6

starting wave is traveling back at velocity $(-c)$. Since the former has a head start of $(-ac\theta)$, the latter overtakes it at the time

$$t_0 = ac\theta/(c-ac) = a\theta/(1-a). \tag{11}$$

The graph at this instant resembles an arrowhead with the peak of the 'barb' at $d=1$, $x=-ct_0$.

Further development could be followed by the 'shear rule,' but may also be followed quantitatively. Introducing the notation shown in Fig. 6 (where $L>0$, and the shear rule has been used to locate the leading edge) we find from (9)

$$U = -dL/dt = c\,(1-y-a),$$

and by similar triangles $(y-\frac{1}{2})/L = 1/2ct.$ (12)

Elimination of y gives a differential equation for L

$$(1/c)\; dL/dt = (a-\tfrac{1}{2}) + L/2ct. \tag{13}$$

The solution of this is (where C is an integration constant)

$$L = C\sqrt{ct} + 2c\,(a-\tfrac{1}{2})\,t.$$

To evaluate the integration constant, we note that when $t=t_0$ given in (11), $L=ct_0$. The result is

$$L = 2c\,[(a - \tfrac{1}{2})\,t + \sqrt{a\theta t\,(1-a)}], \tag{14}$$

and from (12) the peak density $d(-L,t) = y$ is

$$d(-L,t) = a + \sqrt{\theta a\,(1-a)/t}. \tag{15}$$

First note that if $a \geq \frac{1}{2}$ (original density) $\geq \frac{1}{2}$ (maximum density), or traffic so heavy that $V \leq \frac{1}{2}\,c$, then (14) shows that the shock never returns to the point of disturbance ($x=0$) but travels backwards indefinitely though its 'strength,' as measured by the height of the discontinuity, continually decreases. If we say, rather arbitrarily, that the shock is negligible when it causes a velocity change of less than 10 per cent, then it may be said to disappear when $c\,[1 - d(-L,t)] = 0.9\,(1-a)\,c$ which, using (15), will ocur at a time

$$t_f = 100\,\theta a/(1-a).$$

Since $a \geq \frac{1}{2}$ in the case we are discussing, the shock wave 'lasts' for a period at least 100 times as long as the original delay. We note also that the density at $x=0$ is equal to $\frac{1}{2}$ for all $t>0$ so that an observer stationed there would see the traffic become steady almost immediately after the delay.

Secondly, if $a \leq \frac{1}{2}$, the 't' term in equation (14) will be negative and will eventually dominate the '$\sqrt{t}$' term. Therefore the shock wave will eventually reverse its motion and return to the origin at a time determined by $L=0$ which gives

$$\text{time of return} = t_r = \theta a\,(1-a)/(\tfrac{1}{2} - a)^2. \tag{16}$$

At this moment, the fractional density at $x=0$ suddenly drops from the value $\frac{1}{2}$ to the original value a as the shock passes with the traffic. An observer at the origin would mark this as the time when the traffic-jam finally clears away. This time can be quite long if the traffic is at all heavy. For example, if $a = \frac{3}{8}$ corresponding to an original density sufficient to hold the speed down to $(\frac{5}{8})\,c$, then the time required to clear the traffic is $t_c = 15\,\theta$. Thus an accident which halted traffic five minutes would, under these conditions, produce a jam requiring an hour and a quarter to clear. (Under conditions of large density $\geq \frac{1}{2}$ the jam might be said to be so great already that a delay could hardly pile traffic up much more.)

We can use the equations already developed to study the effect of a traffic light on an otherwise steady stream of traffic. Suppose the cycle of the light is that shown in Fig. 7; here we have shifted the origin of time to correspond to the start of a red light. In our previous formulas, we merely replace t by $t - \theta$. Consider first the case in which a is small enough so that the shock wave passes the light ($x=0$) while it is still green. Then the second red light, like the first, will produce the situations shown in Fig. 5, the wave will again clear during the following green light, and the cycle will repeat indefinitely.

Now the largest value of a for which this tidy situation can prevail is that which causes the tail of the shock to cross the light ($x=0$) at the instant $t=T$ when the second red light comes on. This critical value of a is given by setting $L=0$ and (recalling the shift in time-origin) 't' $=T-\theta$ in (14). The result found is that the shock-wave will clear the light if

$$a \leq \tfrac{1}{2}\,(1-\sqrt{\theta/T}). \tag{17}$$

If a is larger than this critical value, each successive shock will leave a 'residue' which was caught by the red light. We can show that these residues build up indefinitely until (as $t \to \infty$) the entire region behind the light has become blocked up to a certain level. To show this, we note that the 'shear rule' shows that during a green light the density at the light is $\tfrac{1}{2}$ (or less after the shock-wave clears), and the traffic flow through the light is therefore, equation (10), no more than ($\tfrac{1}{4}\,cD_{\max}$) at any instant. The number of cars which can get through during each light cycle is therefore

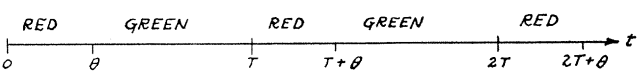

Figure 7

no greater than $\tfrac{1}{4}\,cD_{\max}\,(T-\theta)$. The *average* current which can pass through the light is thus no greater than

$$F_0 = \tfrac{1}{4}\,D_{\max}\,c\,(T-\theta)/T. \tag{18}$$

This may be seen to be precisely the average flow when the shock wave just fails to clear the light. Thus whenever a is *greater* than the critical value given in (17), the average current through the light is limited by the value F_0 given in (18). Now although the instantaneous current (or flow) can vary from place to place, the *average* current over long times must be the same at all points on the road. Therefore, for large a, the current behind the light is limited by (18) and the average density behind the light *must* shift sufficiently to limit the average current to this value. Combining (10) and (18) we find that if $a > \tfrac{1}{2}\,(1-\sqrt{\theta/T})$, the density will change from the value a to an *average* value $\tfrac{1}{2}\,(1+\sqrt{\theta/T})$.

Recapitulating, the light has only a momentary effect on traffic flow as long as the fractional density is less than the critical value $\tfrac{1}{2}\,(1-\sqrt{\theta/T})$ but causes a sudden 'back up' to the average value $\tfrac{1}{2}\,(1+\sqrt{\theta/T})$ as soon as the critical density is exceeded.

For a light with equal lengths of red and green phases, the critical fractional car-density is only about $1/7$ (corresponding to $V = 6/7\, c$), and as soon as this value $1/7$ is exceeded the light backs traffic up to a density of $6/7$ thereby reducing the speed of individual cars by a factor of 6! Even with a light favoring the 'main' road by 9-to-1 the corresponding values are $1/3$ (or $V = 2/3\, c$) and $2/3$ (or $V = 1/3\, c$).

REFERENCES

1. A. Reuschel, *Zeits. d. Oesterreich. Ing. u. Arch. Vereines* **95,** 59 and 73 (1950).
2. A. Reuschel, *Oesterreich. Ing. Archiv* **4,** 193 (1950).
3. L. A. Pipes, *A Proposed Dynamic Analogy of Traffic*, Special Study, Institute of Transportation and Traffic Engineering, University of California, July 11, 1950.
4. L. A. Pipes, *An Operational Analysis of Traffic Dynamics*, Institute of Transportation and Traffic Engineering, University of California (no date).
5. W. Prager, *Proceedings of the Symposium on Operations Research in Business and Industry*, Midwest Research Institute, April 1954.
6. G. F. Newell, *J. Opns. Res. Soc. Am.* **3,** 176 (1955).
7. E. S. Olcott, *J. Opns. Res. Soc. Am.* **3,** 147 (1955), especially Figs. 1–3.